

MANINGRIDA, Australia

WMO: 951420

Lat: 12.0569S

Lon: 134.2339E

Elev: 30

StdP: 100.97

Time Zone: 9.50 (AUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	17.6	6.9	6.2	24.9	8.8	7.0	24.5	8.6	26.3	8.2	26.8	2.5	170	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.7	33.8	25.1	33.3	25.1	32.8	25.0	27.9	31.2	27.2	30.7	26.8	30.4	4.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	29.4	26.4	22.0	29.0	25.9	21.4	28.7	89.5	31.4	86.6	30.8	84.7	30.4	34.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	7.2	6.5	DB	13.1	35.6	1.7	0.6	12.0	36.1	11.0	36.4	10.1	36.7	8.9	37.1	(4)
(5)			WB	10.7	29.1	1.5	1.9	9.6	30.5	8.7	31.6	7.9	32.7	6.8	34.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	28.4	28.3	28.0	27.6	26.6	24.8	24.4	24.3	25.8	27.5	28.8	29.0	(6)	
	DBStd	2.10	1.20	1.17	1.09	1.00	1.37	1.92	1.63	1.46	1.31	1.14	0.83	1.12	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6184	570	513	557	527	515	444	445	444	475	541	563	590	(10)	
	CDD18.3	3143	312	279	298	277	257	195	187	186	225	283	313	332	(11)	
	CDH23.3	33892	3572	3152	3191	2909	2494	1600	1518	1655	2261	3292	3977	4270	(12)	
	CDH26.7	12987	1372	1188	1138	1088	872	474	444	516	784	1337	1812	1962	(13)	
(14) Wind	WSAvg	3.3	3.0	3.0	2.8	3.1	3.6	4.0	4.0	3.5	3.2	3.2	3.2	3.1	(14)	
(15) Precipitation	PrecAvg	1284	281	276	293	126	23	1	1	0	1	13	55	224	(15)	
	PrecMax	2174	707	552	1224	735	203	9	11	2	11	71	290	603	(16)	
	PrecMin	689	63	44	72	8	0	0	0	0	0	0	0	13	(17)	
	PrecStd	377	131	120	211	128	40	2	3	0	2	19	66	157	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	33.9	34.1	33.5	32.9	31.8	31.8	32.2	33.3	34.0	34.0	34.7	(19)	
		MCWB	26.6	26.1	25.7	23.8	23.5	22.1	20.6	19.5	20.0	22.1	25.0	26.1	(20)	
	2%	DB	33.1	32.9	32.7	32.7	32.0	31.0	30.8	31.2	32.0	32.7	33.2	33.7	(21)	
		MCWB	26.5	26.3	25.6	23.9	23.4	21.9	20.7	19.8	21.7	23.7	25.0	25.8	(22)	
	5%	DB	32.3	32.2	31.8	32.0	31.3	30.2	30.1	30.3	31.1	32.1	32.7	33.1	(23)	
		MCWB	26.3	26.2	25.7	24.1	23.1	21.6	20.7	19.9	21.9	23.9	25.0	25.8	(24)	
	10%	DB	31.5	31.4	31.1	31.3	30.7	29.4	29.2	29.5	30.4	31.5	32.2	32.5	(25)	
		MCWB	26.2	26.0	25.7	24.2	22.9	21.3	20.4	20.0	22.0	23.8	24.9	25.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.5	28.0	28.5	26.9	26.2	24.4	23.7	23.7	25.0	26.2	27.1	28.3	(27)	
		MCDB	31.3	31.5	30.8	30.1	29.7	28.7	28.2	28.2	29.0	30.8	31.7	32.0	(28)	
	2%	WB	28.0	27.2	27.4	26.3	25.6	23.8	23.1	23.1	24.4	25.5	26.5	27.3	(29)	
		MCDB	31.3	30.8	30.3	29.8	29.1	28.2	27.3	27.5	28.6	30.3	31.2	31.5	(30)	
	5%	WB	27.2	26.8	26.8	25.9	25.1	23.2	22.6	22.6	23.9	25.0	26.0	26.8	(31)	
		MCDB	30.4	30.4	29.8	29.4	28.5	27.7	27.0	27.0	28.4	29.9	30.9	31.1	(32)	
	10%	WB	26.7	26.4	26.4	25.5	24.5	22.7	22.0	22.1	23.5	24.6	25.6	26.3	(33)	
		MCDB	29.9	30.0	29.4	29.0	28.2	27.2	26.7	26.5	28.0	29.5	30.5	30.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.2	6.4	7.7	8.8	9.7	10.6	11.4	10.2	8.7	7.6	6.7	(35)	
		MCDBR	7.4	7.2	7.5	8.8	9.2	9.7	10.9	12.1	10.9	9.1	7.9	7.6	(36)	
	5% WB	MCWBR	3.1	3.0	2.8	2.7	3.0	3.4	3.9	4.9	4.1	3.1	2.5	2.5	(37)	
		MCWBR	6.5	6.6	6.6	7.1	7.8	8.7	9.4	9.8	8.7	8.2	7.4	6.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.411	0.402	0.391	0.379	0.372	0.357	0.350	0.360	0.407	0.426	0.426	0.408	(40)	
		taud	2.531	2.553	2.586	2.599	2.591	2.595	2.604	2.552	2.409	2.380	2.424	2.522	(41)	
	Edn at Noon		932	934	927	907	883	881	896	913	898	902	912	934	(42)	
		Edh at Noon	112	109	103	96	92	89	90	99	121	128	123	112	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.32	5.48	5.37	5.48	5.22	5.00	5.27	6.03	6.68	6.97	6.81	5.93	(44)	
	RadStd		0.64	0.79	0.61	0.64	0.36	0.23	0.19	0.20	0.25	0.29	0.28	0.56	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.40	N/A	N/A	N/A	N/A	-0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page